

Gold Futures Spread Intelligence

One metal, four contracts: do India's gold futures agree on the price of gold, and can the gaps be traded after real costs?

130 MCX contracts analysed (10,869 daily records)	0 unexplained data errors after integrity checks	60-75 bp extra price of the 8 g coin over GOLDM, 2024 to 2026	about 2.7% average premium during the Dec 2025 – Mar 2026 crash
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Status as of 3 October 2026. All numbers come from official MCX Bhavcopy files and are reproducible with the scripts in our repository. Nothing in this report is estimated or invented unless it is clearly marked *projection* (Section 9).

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1. Summary in one page

India's commodity exchange, MCX, lists four gold futures contracts that differ only in size and purity. Once their prices are converted to the same unit, rupees per gram of pure gold, they should be nearly identical. We checked that with 130 contracts of official exchange data (October 2023 to September 2026), measured every gap, and tested two trading strategies on data they had never seen, after realistic costs.

What we found

- **The conversion works.** In normal months the 100 g bar (GOLDM) and the 10 g bar (GOLDTEN) are priced the same per pure gram: +0.8 basis points, with a 95% range of −2.4 to +4.1. A mistake in our purity correction would show up as a 40-point gap.
- **The coin always costs more.** The 8 g coin (GOLDGUINEA) was about 75 bp (0.75%) dearer per gram than GOLDM in 2024 and about 60 bp in 2025–26. It was dearer in all 34 contract cycles we checked, so it is a lasting premium, not a passing mistake.
- **The 1 g contract changed sides.** GOLDPETAL traded about 140 bp (1.4%) *below* GOLDM through 2024, then about 64 bp above it in 2025–26. A premium that can flip is not something to rely on.
- **In a crash the gap explodes.** During the gold sell-off of December 2025 to March 2026, small contracts traded about 2.7% above GOLDM on average, with a monthly peak of 3.8%. The most traded contract moved first and furthest; the small ones lagged.
- **Trading the gaps is not reliably profitable yet.** Our simple strategy made money in the sealed test (+₹1.9 lakh at 5 bp slippage), but all of it came from January 2026; in the other months it lost ₹77,410. Our fair-price strategy made +₹12,990, but its 95% range (−₹18,671 to +₹50,806) includes zero.
- **The answer is honest by design.** Every rule was fixed and time-stamped in Git before the test data was downloaded. Two more sealed tests (2020–2023 and 2016–2019) are declared and waiting.

Why this matters

Most backtests look profitable because they are tuned on the same data they are judged on, or they ignore costs. We did neither. The result is a map of when these gold contracts disagree, how large the disagreement is, and the conditions under which it can be traded: in sharp market falls, not in quiet months.

2. The problem, in plain words

A **futures contract** is an agreement to buy or sell something at a fixed price on a future date. On MCX, people trade gold futures every day; the price moves with the gold price. Gold is sold in many sizes, so MCX offers four gold contracts for different budgets:

Contract	Size of one lot	Price is quoted per	Purity	Expires around	Gold traded per day*
GOLDM (Gold Mini)	100 g	10 g	99.5%	5th of the month	479 kg
GOLDTEN	10 g	10 g	99.9%	last day of month	34 kg
GOLDGUINEA	8 g (coin)	8 g	99.9%	last day of month	8 kg
GOLDPETAL	1 g	1 g	99.9%	last day of month	12 kg

** Median over all our data. GOLDM trades about 15–60 times more gold than the others. GOLDTEN only started in April 2025, which makes it the most reliable price.*

Because each contract quotes its price for a different amount and purity of gold, the raw prices look very different (for example ₹97,214, ₹97,428, ₹78,086 and ₹9,820 on the same day).

Underneath, they are all the same metal. The hackathon problem asks us to:

- **Put all four on a common basis**, so the prices can be compared fairly;
- **measure the gaps** between them: how big, how stable, and when they change;
- **test whether any gap can be traded** for profit after real costs (fees, taxes, and the cost of not getting the exact price);
- **present the analysis honestly**, without the shortcuts that make backtests look better than reality.

Why would the prices differ at all?

Several real reasons can make the same gold cost slightly different amounts per gram in different contracts: smaller pieces cost more to mint and deliver per gram; small contracts trade less, so their prices update more slowly; and the contracts expire on different dates, so money tied up for longer costs interest (the *cost of carry*). Part of our job is to separate these normal, explainable gaps from gaps that are unusual and might be tradeable.

The brief, point by point

The brief asks for	Where this report answers it
Normalize size, quote unit and purity; spot unusually cheap or dear contracts	Sections 5 and 7; Strategy B in Section 8
Analyze the futures curve; separate roll-down from genuine curve changes	Section 7, roll-down vs curve move
Walk-forward backtest with no look-ahead, costs and thin-day liquidity	Sections 5, 6 and 8
Dashboards or alerts that stay quiet without a signal; attribute performance to the strategy, not gold	The dashboard (Section 11); Section 8, where the profit came from
Track listing, liquidity, tender periods and expiry; keep every entry and exit inside the contract calendar	Section 8, contract calendar check; Section 4
Validate on unseen data; report after costs on the contracts held	Sections 6 and 8

3. Words you will see

Term	Meaning in this report
MCX	Multi Commodity Exchange of India, where these gold futures trade.
Bhavcopy	The exchange's official daily record of each contract: open, high, low, close, volume, value traded.
Expiry	The date a futures contract ends. Each contract month is a separate contract (a <i>cycle</i>).
Per pure gram	Price divided by quote size and purity. Our common unit: ₹ per gram of 100% gold.
Basis point (bp)	One hundredth of a percent. 100 bp = 1%. A 60 bp gap on ₹10,000 gold is ₹60 per gram.
Spread / gap / premium	How much more one contract costs than another, per pure gram, in bp.
Cost of carry	Interest-like cost that makes a later-expiring contract slightly dearer. We measure it from the data.
Settlement close	The exchange's official end-of-day price. Not always a price anyone actually traded at.
VWAP	Average traded price of the day = total value traded ÷ grams traded. A realistic fill price.
Slippage	Getting a slightly worse price than planned. We test 0, 5 and 10 bp per trade leg.
Backtest	Replaying a trading rule on past data to see what it would have earned.
Out-of-sample / sealed test	Data the rule never saw while it was being designed. The only fair test.
95% range	The range the true value most likely lies in, given how noisy the data is. If it includes zero, the result could be luck.
Lakh	₹1,00,000.

4. Our data and how we checked it

Source. MCX Bhavcopy, downloaded by hand one contract at a time from mcxindia.com (the site blocks automated downloads). We do not use stitched 'continuous' price series from charting sites, because joining different contracts together creates false jumps.

130 contracts	10,869 daily records	10 Oct 2023 first date	30 Sep 2026 last date
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Every file is identified by its contents (not its file name), duplicates are removed automatically, and each row is checked before it is used:

Check	What it catches	Rows failing
Duplicate rows	Same contract and date twice	0.00%
Row after expiry	Wrong dates	0.00%
Open or close outside the day's high-low	Corrupt prices	0.00%
Zero or negative price	Corrupt prices	0.00%

Check	What it catches	Rows failing
Missing trading day inside a contract's life	Incomplete downloads	0.00%
Weekend date that is not a known special session	Wrong date returned	0.00% (3 special sessions found and expected)
Previous close $\neq$ last row's close	Broken history	1.31% (141 rows), all right after a no-trade day; explained, flagged and excluded
Price > 5% away from same-month peers	Wrong units or lot sizes	0.10% (11 rows): 10 GOLDM rows from 21 Jan to 1 Feb 2026, and GOLDGUINEA on its thin expiry day, 31 May 2024

The flagged prices are real. On 30 January 2026 gold fell about 10% in a day; GOLDM settled at its day's low (its lower price limit) while the smaller contracts settled higher. 1 February 2026 was a Sunday Budget-day special session. News reports confirm the crash (Section 11). We kept these rows and did not change any rule because of them.

Sealed data is kept apart. 37 contracts from 2020 have already been downloaded for the next sealed test. The loading script writes them to a separate file that no analysis reads; only their names and dates were checked.

How far is the official close from a real trade?

We compared each day's settlement close with that day's average traded price (VWAP). For a single contract the median gap is 17–26 bp. For the *gap between two contracts*, which is what we trade, the median is only 6.8–8.5 bp because the day's gold move cancels out. This is why we assume 5 bp of slippage per leg as the realistic case, and also show 0 and 10 bp.

5. How we solve it, step by step

Step 1: Convert every price to rupees per gram of pure gold

price per pure gram = settlement price $\div$ grams per quote $\div$ purity

15 Jul 2025	Official price	$\div$ grams	$\div$ purity	₹ per pure gram
GOLDM (Aug expiry)	97,214	10	0.995	9,770.25
GOLDTEN (Jul expiry)	97,428	10	0.999	9,752.55
GOLDGUINEA (Jul expiry)	78,086	8	0.999	9,770.52
GOLDPETAL (Jul expiry)	9,820	1	0.999	9,829.83

Four very different prices become four almost equal ones. The remaining differences, a few tenths of a percent, are what we study.

Step 2: Line up the expiry dates

GOLDTEN, GOLDGUINEA and GOLDPETAL expire on the same day, so they can be compared directly. GOLDM expires about five days later, and a later contract naturally costs a little more (cost of carry). We measure that carry every day from the data itself, by comparing neighbouring expiries of the same contract, and adjust GOLDM to its partner's expiry date. The adjustment is small (median about 13 bp) but without it GOLDM would look slightly too expensive.

Step 3: Measure the gaps

For every pair of contracts and every day we compute the gap in basis points: $10,000 \times \ln(\text{price A} \div \text{price B})$. We then average by contract cycle and compute 95% ranges with a *week-block bootstrap*: we resample whole weeks rather than single days, because one day's gap is strongly linked to the next, and treating them as independent would make us look more certain than we are.

Step 4: Turn gaps into trading rules

We tested two families of rules. Both buy the contract that looks cheap and sell the one that looks expensive in equal grams of gold, so a rise or fall in the gold price itself roughly cancels out. They profit only if the gap returns to normal.

	Strategy A: same-expiry z-score	Strategy B: GOLDM-anchored fair price
Idea	If today's gap between two small contracts is unusually far from its recent average, bet that it returns.	GOLDM is the most traded, so trust its price. Fair price of a small contract = GOLDM + that contract's usual premium. Trade when the small contract strays from its fair price.
'Unusual' means	More than 2 standard deviations from the average of the previous 10 days	More than 1.5 standard deviations from its average premium on all previous days
Size	40 g per leg	200 g per leg (2 GOLDM lots vs 20 / 25 / 200 lots)
Fills at	Next day's settlement close	Next day's average traded price (VWAP)
Exit	Gap back to normal, or just before expiry	Premium back to usual, or 5 business days before expiry
Research basis	Gatev, Goetzmann & Rouwenhorst (2006)	Built from our own measurement that GOLDM = GOLDTEN

We also compared research-backed alternatives (Ornstein-Uhlenbeck mean-reversion scores, a cost filter, and a Kalman filter that treats thin-market prices as noisy). None made money on the training data after costs, so none was promoted.

Step 5: Charge realistic costs

Cost item	Rate used	Source / status
MCX transaction fee	₹2.10 per lakh of turnover	MCX filing, effective 1 Oct 2024
Commodity transaction tax	0.01% on the sell side	Non-agricultural futures rate
Stamp duty	0.002% on the buy side	Broker charge sheets (assumption)
SEBI fee	₹10 per crore	Assumption
Brokerage	₹20 per order, max 0.03%	Discount broker (assumption)
GST	18% on brokerage and fees	Assumption
Slippage	0 / 5 / 10 bp per leg per side	5 bp = realistic case (Section 4)

Each trade has four legs (buy and sell on entry and exit), so every cost is paid four times.

6. How we keep the test honest

The most common way to fool yourself in trading research is to tune a rule until it looks good on the past, and then report that same past as the result. We split the data **by calendar date** into periods with different jobs, and wrote down every rule, and the exact test dates, **before** the test data existed on our computers. Git commit timestamps prove the order.

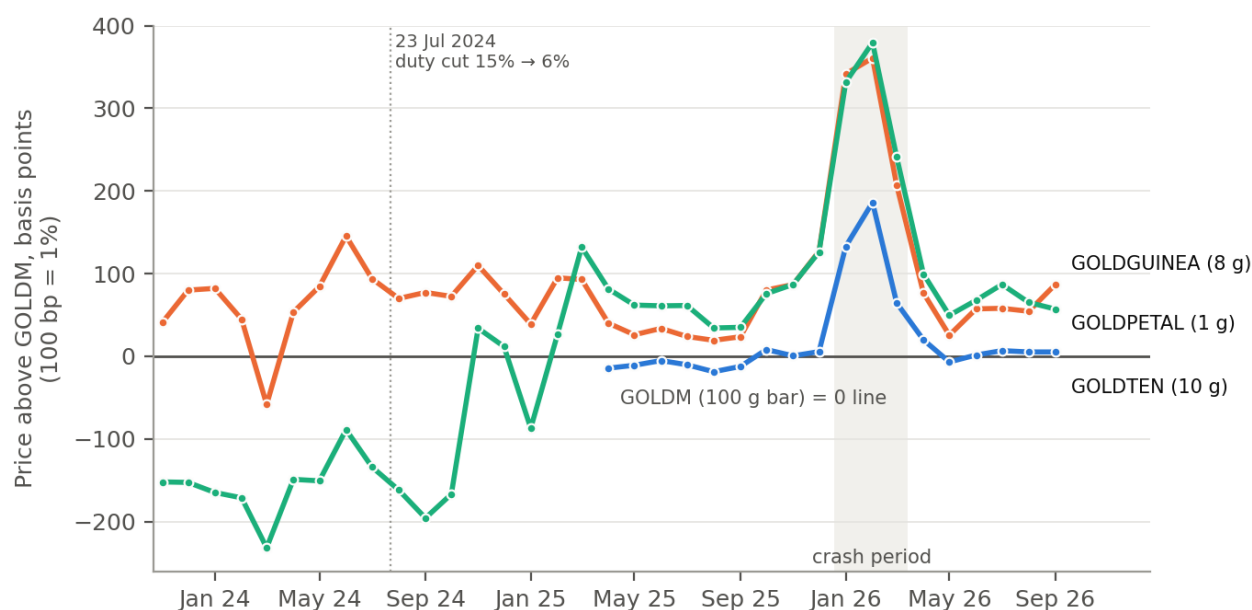
Period	Dates	Used for
Train / Development	up to 30 Apr 2025 (A); up to 5 Aug 2025 (B)	Designing rules and choosing settings by a pre-declared rule
Test 1 (Strategy A only)	1 May – 31 Jul 2025	First sealed test, run once
Sealed holdout	Aug 2025 – May 2026	Final test of both strategies, run once
Sealed test 2	1 Jan 2020 – 9 Oct 2023	Declared before download; run once on the improved rules
Sealed test 3	1 Jan 2016 – 31 Dec 2019	Declared before download; run once on the improved rules

Time (IST)	Commit	What happened
1 Oct 2026, 21:45	e5e0a21	Holdout window declared, download list written
1 Oct 2026, 22:15	045ab2b	Strategy B rules and setting ($k = 1.5$) frozen
1 Oct 2026, 23:01	7acf18e	Holdout data (39 MCX files) uploaded
1 Oct 2026, 23:05	3e9e82d	Holdout run once; results recorded as they came out
2 Oct 2026, 03:28	35d7e7b	Sealed test 2 (2020–2023) declared before download
3 Oct 2026, 01:13	a8cda54	Sealed test 3 (2016–2019) declared before download
3 Oct 2026, 01:47	8f16845	Test 2 shortened to end 9 Oct 2023: the 2024 files included late-2023 days, now seen; sealed rows walled off in code
3 Oct 2026, 02:43	79129fc	2020 files arrive and go straight to the sealed file, unread

Other safeguards

- Signals use only information available at that day's close; trades happen on the next day.
- One position per contract type at a time, so one market move is not counted as several independent wins.
- Positions close before MCX's compulsory delivery period at expiry.
- We report results at three slippage levels, the 95% range, and how concentrated the profit is.
- Weekend dates must be known special sessions (Diwali Muhurat 2023, Budget days 2025 and 2026); dates are read in a fixed format so day and month cannot swap.
- A bug found during development (the 10-day average included the current day) was fixed *before* any test was run, and is logged.
- An earlier AI-generated draft from another tool produced statistics for contracts that were not in our data and references that did not exist. We discarded all of it and rebuilt every number from the raw files; every reference in Section 11 was checked.

7. Results: what the gaps look like



Monthly average price of each small contract above GOLDM, per pure gram, after the carry adjustment. GOLDTEN starts in April 2025, when MCX listed it.

The chart shows three different worlds. In 2024 the coin sat above GOLDM while the 1 g contract sat well below it. Through the quiet months of 2025–26 GOLDTEN sits on GOLDM's line and the coin and 1 g contracts sit together about 20–130 bp above it. From December 2025 to March 2026, while gold crashed, every small contract became much dearer than GOLDM, then the gaps returned to normal by April–May 2026.

Gap (A minus B, per pure gram)	2024	2025–26 quiet months	Dec 2025 – Mar 2026
GOLDM – GOLDTEN	not listed yet	+0.8 bp (–2.4 to +4.1)	–100.5 bp (–153.9 to –54.0)
GOLDM – GOLDGUINEA	–74.9 bp (–88.9 to –60.1)	–59.5 bp (–67.1 to –51.7)	–263.3 bp (–329.5 to –200.5)
GOLDM – GOLDPETAL	+140.3 bp (+122.5 to +155.9)	–63.6 bp (–73.2 to –52.4)	–269.4 bp (–331.2 to –209.5)
GOLDTEN – GOLDGUINEA	not listed yet	–59.3 bp (–65.3 to –53.1)	–166.6 bp (–195.8 to –137.9)
GOLDTEN – GOLDPETAL	not listed yet	–70.8 bp (–75.4 to –66.2)	–169.2 bp (–193.8 to –145.0)
GOLDGUINEA – GOLDPETAL	+211.1 bp (+199.4 to +221.7)	–2.2 bp (–10.7 to +7.3)	–4.1 bp (–17.8 to +11.8)

Brackets show the 95% range. 2024: 60–65 weeks; 2025–26 quiet months: 61–75 weeks; crash period: 18 weeks. The crash period was chosen after looking at the chart, so that split describes the data rather than testing a prediction.

Roll-down or a real move?

A futures price drifts toward the spot price as expiry approaches, even if nothing in the market changes. This is *roll-down*, and it follows directly from the cost of carry. For each month we split the price change of the most-traded contract into roll-down (what that day's carry implies for the days that passed) and the curve move (everything else, mostly gold itself).

Contract	Months	Roll-down per month	Curve move per month (average size)	Roll-down share
GOLDM	32	−0.59%	±4.43%	12%
GOLDTEN	17	−0.96%	±5.48%	15%
GOLDGUINEA	32	−0.73%	±4.65%	14%
GOLDPETAL	33	−0.77%	±4.51%	15%

We also fit a straight line through every listed expiry of each contract type every day; its slope is the annualised carry across the whole curve. It was about 4.5–7% a year through 2025, rose to 14–18% from December 2025 to June 2026, and fell back to 7–9% from July 2026. All four contract types, fitted separately, moved together, so this is market-wide. A straight line fits each day's curve to within about 4–5 bp.

Roll-down is small and predictable; the curve move is large and is where the risk sits. Measured carry for GOLDM was about 4–6% a year in 2023–25 and about 15% in 2026; the 2026 level is what the data shows, and we have not found its cause.

What this tells us

- **The purity correction is right.** If it were wrong, GOLDM and GOLDTEN would differ by about 40 bp in every month; they differ by less than 1.
- **The coin premium is structural.** GOLDGUINEA was dearer than GOLDM in all 34 contract cycles from 2024 to 2026, so the gap does not close and cannot be captured by holding to expiry. A likely reason, which we have not proved, is the higher cost of minting and delivering coins.
- **The 1 g premium is not.** GOLDPETAL was below GOLDM in 2024 and above it from early 2025; it agrees with GOLDM's sign in only 21 of 35 cycles. We checked the raw files: the units and lot values are right, so the flip is real. Its cause is not yet known.
- **Stress reveals who leads.** In a fast fall, GOLDM (where most gold trades) reprices first; the small contracts follow later. That delay is the only place we found real money (Section 8).

8. Results: can the gaps be traded?

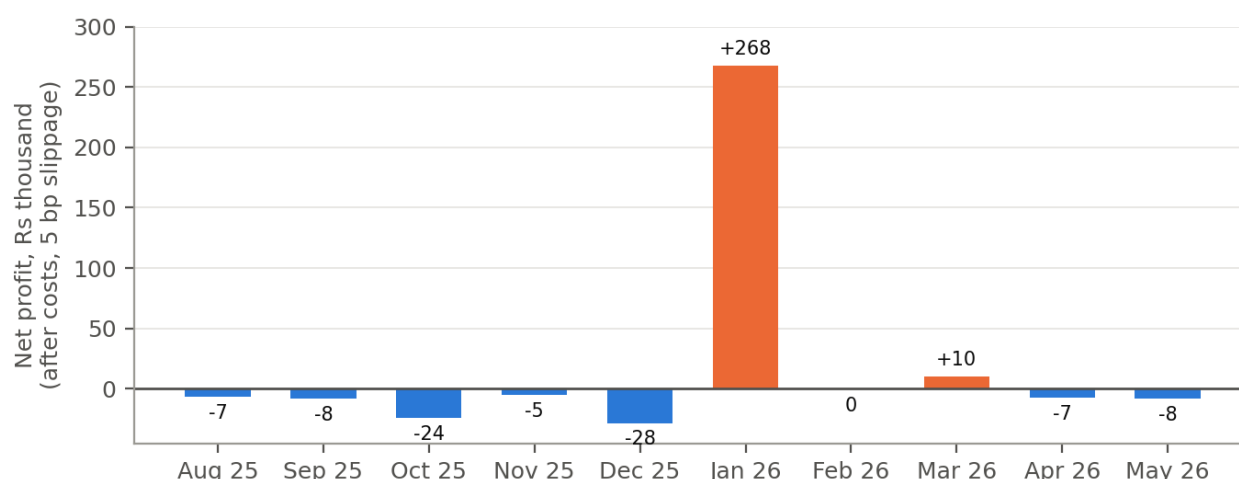
Training and first test (Strategy A, 2025)

Period	Trades	Net at 0 bp	Net at 5 bp	Net at 10 bp
Train (to Apr 2025)	9	lost money even before costs (gross −₹3,939)		
Test 1 (May–Jul 2025)	19	+₹10,292	−₹4,466	−₹19,224

Break-even slippage was about 3.5 bp: profitable only with near-perfect fills. 95% range at 5 bp: −₹9,849 to +₹2,350.

Sealed holdout, Aug 2025 – May 2026 (run once)

Strategy	Trades	Net at 0 bp	Net at 5 bp	Net at 10 bp	95% range at 5 bp
A: same-expiry z-score (40 g)	156	+₹3,65,755	+₹1,90,340	+₹14,926	−₹97,634 to +₹6,19,835
B: GOLDM fair price (200 g)	12	+₹74,808	+₹12,990	−₹48,828	−₹18,671 to +₹50,806



Strategy A in the sealed holdout, net profit by month of entry (₹ thousand, after costs, 5 bp slippage).

How to read these numbers

- **Strategy A's profit is one event.** 30 trades in January 2026 made +₹2,67,750. The other 126 trades lost ₹77,410 and won only 27% of the time. The top five January trades alone are 54% of the total. Repricing every trade at the day's average traded price instead of the settlement close still gives +₹1,68,114, so the January profit is not just a settlement-price illusion.
- **Strategy B works only where a real premium exists.** It made +₹27,799 on GOLDS GUINEA and +₹6,877 on GOLDPETAL, but lost ₹21,686 on GOLDTEN, where there is no premium and the strategy trades noise. It used only 6 independent weeks, so its 95% range includes zero.
- **Costs decide everything.** Both strategies are clearly profitable at 0 bp slippage and lose at 10 bp. Realistic execution, 5 bp, sits right on the edge.

Where the profit came from: the gap or gold?

Both legs of every trade hold the same grams, so gold's own move mostly cancels. It does not cancel completely: a premium still open at exit, and GOLDM's 99.5% purity against 99.9%, leave a small exposure. We split every trade's profit exactly into a **gap part** (the gap changing, valued at the entry gold price) and a **gold part** (gold's move times the gap still open). The split reproduces every recorded trade profit to the rupee.

Run	Trades	Gross profit	From the gap	From gold's move	Correlation with gold's move
A · test 1	19	₹14,624	₹14,847	−₹223	−0.29
A · sealed holdout	156	₹4,10,655	₹4,04,944	₹5,710 (1.4%)	0.55
B · development	8	₹1,13,653	₹1,13,730	−₹77	0.57
B · sealed holdout	12	₹86,572	₹90,739	−₹4,167 (−4.8%)	−0.04

Strategy A's correlation of 0.55 comes from January 2026: gold rose about 7.8% during those trades while the gaps widened and closed. That month the gap part made ₹3,08,106 and gold's move ₹9,997. Outside January the correlation is 0.02. The profit came from the gap, not from betting on gold's direction.

Do the alerts mean anything?

Using the same fair-price model, we listed every past alert (68 episodes since December 2023) and what the gap did next. An alert does *not* make the gap more likely to close: it halved within 10 trading days 52% of the time on alert days and 56% on ordinary days. But alerts mark the large gaps: the median gap was 180 bp and it closed by a median 47 bp within 10 days, against 3 bp on ordinary days. A round trip costs about 24 bp at 5 bp slippage, so alerts clear costs at the median and only barely on average (30 bp). Most alerts during the crash did not close within 10 days.

Did every trade fit inside its contract?

MCX gold contracts enter a 5-business-day tender (delivery) period before expiry, and brokers close positions before it, for example by 29 January 2026 for the 5 February 2026 expiry. We checked every trade: all started after their contract was trading, and every Strategy B trade closed before the tender period. Strategy A's rule closed positions 3 calendar days before expiry, which is too late: 17 of its 156 holdout trades (and 3 of 19 in test 1) were held 1–3 business days into the tender period, where a broker would not allow them.

Strategy A, net at 5 bp	All trades	Trades a broker would allow
Sealed holdout	+₹1,90,340 (156)	+₹1,53,382 (139)
Test 1	−₹4,466 (19)	−₹1,922 (16)

The sealed runs are not repeated; this is reported as a finding. The improved rules for the next sealed tests use the 5-business-day exit.

Our verdict today. The gaps between India's gold contracts are real, measurable and explainable. Trading them in quiet markets does not pay after costs. Trading them during sharp crashes did pay in our data, because small contracts lag GOLDM. With only one such crash inside our sealed test, this is a promising lead, not a proven edge, and the next data round is designed to test exactly this.

9. What more data will add (with projections)

GOLDGUINEA has traded on MCX since 2008 and GOLDPETAL since 2011, so years of the same official data exist. GOLDTEN only started on 1 April 2025, so its history can only grow month by month.

Batch	Contracts	Files	Status	What it adds
2024	GUINEA, PETAL, GOLDM, all 2024 expiries	36	Done	Development data; showed the 1 g flip; includes the 23 Jul 2024 duty cut
2025–26 gaps	GOLDM May–Sep 2026, small contracts Jun–Sep 2026, PETAL Jan 2025	16	Done	Complete 2025–26 coverage
Gap	GOLDTEN Aug 2026	1	Missing	Last 2025–26 gap
2020	Same three, all 2020 expiries	37	Done, sealed	Sealed test 2: Mar 2020 COVID crash and Aug 2020 peak
2021–23	Same three, Feb 2021 – Dec 2023 expiries	107	To do	Rest of sealed test 2
2016–19	Same three, all 2016–2019 expiries	144	To do	Sealed test 3

Projected precision

The 95% range of an average shrinks roughly with the square root of the number of independent weeks. Assuming new data is about as variable as today's (an assumption, not a guarantee):

Measurement	Today	After the sealed tests
Strategy B, average profit per trade	Range from 6 independent weeks	About 40% as wide if 2016–2023 trades at a similar pace
Crash episodes to test the 'lag in a crash' idea	1 in a sealed test (Jan 2026)	Several more in 2016–2023, judged on rules frozen beforehand
Years in which the coin and 1 g premiums are measured	3 (2024, 2025, 2026)	Up to 11 (2016–2026), showing whether the 1 g flip has happened before
Quiet-month premium margin, coin and 1 g (2025–26)	± 7.7 and ± 10.4 bp (74 weeks)	Narrows only as new months arrive; sealed years are reported separately

Projection: computed from today's measured variability; actual ranges will be known only after the data arrives.

Planned improvements, to be fixed before the next sealed test

- **Price-limit filter:** skip days when a contract settles at its daily limit (close equal to the day's low or high on a large move), because nobody can reliably trade at that price. Detectable from the Bhavcopy alone.
- **Faster-adapting 'usual premium':** Strategy B's average of all past days lagged the 2026 spike, so it held some trades for over 100 days. A recent-weeks average or a Kalman filter will be compared on development data only.
- **Stress detector:** trade only when GOLDM has moved sharply, since that is where the lag appeared. This rule will be written down before we see 2020–2023, and judged only there.
- **Exit before the tender period in every strategy:** 5 business days before expiry, as Strategy B already does.
- **Drop GOLDTEN from Strategy B:** no premium to trade, as both our measurements and the holdout show.
- **Treat the 1 g contract with care:** its premium flipped sign in early 2025, so a fair-price rule built on one year's 'usual premium' can be badly wrong in another. Rules will be built on 2024–26 together and judged on the sealed years.
- **Independent cross-checks:** IBJA's published daily 999 and 995 gold rates (recent days only) to verify our purity ratio and carry estimate against physical gold.

What we will report after each batch: the same tables as this report, updated; whether the sealed-test result clears zero at 5 bp; and whether profits still come from crashes only.

10. Limitations

- **Short working history.** Three years of development data and one large crash. Strong conclusions about trading need several crashes; the sealed tests in Section 9 address this.
- **Daily data only.** We see one official close and one average price per day, not every trade. Real fills could be better or worse than our 5 bp assumption, especially in a crash, when prices move fastest.
- **Some costs are assumptions.** Stamp duty, SEBI fee, brokerage and GST rates come from typical broker charge sheets.
- **Liquidity limits size.** GOLDDUINEA trades about 8 kg a day; larger positions than ours would move the price.

- **Why the premiums exist is a hypothesis.** Minting and delivery costs fit the coin premium, but we have not verified them from MCX's delivery records, and nothing we have explains why the 1 g premium flipped in early 2025.
- **Strategy results predate the 2024 data.** Section 8's numbers come from the single sealed runs of 1 October 2026 and are not re-run.
- **Holidays in the tender check.** The 5-business-day rule counts Monday to Friday and ignores exchange holidays, so it can be off by a day around a holiday.
- **Gaps in data.** GOLDTEN expiring August 2026 has not been downloaded.

11. References and files

Research papers (DOIs checked against Crossref or RePEc)

- Gatev, Goetzmann & Rouwenhorst (2006). Pairs Trading: Performance of a Relative-Value Arbitrage Rule. *Review of Financial Studies* 19(3). doi:10.1093/rfs/hhj020
- Elliott, van der Hoek & Malcolm (2005). Pairs trading. *Quantitative Finance* 5(3). doi:10.1080/14697680500149370
- Avellaneda & Lee (2010). Statistical arbitrage in the US equities market. *Quantitative Finance* 10(7). doi:10.1080/14697680903124632
- Bertram (2010). Analytic solutions for optimal statistical arbitrage trading. *Physica A* 389(11). doi:10.1016/j.physa.2010.01.045
- Leung & Li (2015). Optimal mean reversion trading with transaction costs and stop-loss exit. *IJTAF* 18(3). doi:10.1142/S021902491550020X
- Politis & Romano (1994). The Stationary Bootstrap. *JASA* 89(428). doi:10.1080/01621459.1994.10476870
- Do & Faff (2010). Does Simple Pairs Trading Still Work? *Financial Analysts Journal* 66(4). doi:10.2469/faj.v66.n4.1
- Krauss (2017). Statistical Arbitrage Pairs Trading Strategies: Review and Outlook. *Journal of Economic Surveys* 31(2). doi:10.1111/joes.12153
- Pavabutr & Chaihetphon (2010). Price discovery in the Indian gold futures market. *Journal of Economics and Finance* 34(4). doi:10.1007/s12197-008-9068-9

Data and news sources

- MCX Bhavcopy, mcxindia.com (Market Data → Bhavcopy): all price data in this report.
- MCX to start 8 gm gold coin futures, *Business Standard*, May 2008 (GOLDGUINEA launch).
- 10 gram gold futures contract available from today, *Outlook Money*, 1 Apr 2025 (GOLDTEN launch and specification).
- Gold and silver prices fall after Budget cuts customs duty to 6%, *Business Today*, 23 Jul 2024.
- Government raises import duty on gold and silver to 15%, *StudyCafe*, May 2026 (Notification 16/2026-Customs).
- Silver crashes ₹1.10 lakh, gold down ₹20,000 on MCX, *News Arena*, 30 Jan 2026; Motilal Oswal market note, 30 Jan 2026.
- IBJA daily rates, ibjarates.com (planned cross-check).

Code (repository folder commodity/)

Script	What it does
ingest.py	Reads every MCX file, removes duplicates, validates rows, converts to ₹ per pure gram
testbed.py	Data error rates, settlement-vs-traded price gaps, 95% ranges
accuracy.py	Cost of carry, all six contract pairs, precision of each contract
backtest.py	Strategy A: train, test and holdout, each run once
methods.py	Research-backed alternatives compared on training data
fairvalue.py	Strategy B: development and sealed holdout
uncertainty.py	Week-block bootstrap for the profit ranges
attribution.py	Splits every trade's profit into the gap part and gold's own move
rolldown.py	Splits each month's price change into roll-down and curve move
lifecycle.py	Liquidity build-up, tender periods, every trade checked against its contract
alerts.py	Every past alert and what the gap did next, against ordinary days
curve.py	Daily slope of the whole futures curve per contract type
dashboard/	Interactive website built from the same results (export.py, build.py)